Anxiety-Aware Requirements Engineering: A Vision Paper

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Abstract

Anxiety is referred to as the illness of our century, affecting one in three adults over their lifetime and becoming increasingly prevalent among young people. However, anxiety-related considerations remain largely absent from software development outside specialized domains such as digital health. This gap is particularly critical in requirements engineering, where early design decisions shape user experiences. This vision paper proposes a theoretical framework to prevent anxiety and emotional harm, enabling software engineers, regardless of their mental health expertise, to design systems with improved emotional impact.

CCS Concepts

• **Software and its engineering → Designing software**; **Software prototyping**; • **Human-centered computing → Human computer interaction (HCI)**.

Keywords

Requirements engineering, anxiety, sensors, human-centered design, emotional risks.

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1 The problem: lack of anxiety considerations in requirements engineering

Anxiety disorders are the most prevalent mental health conditions worldwide, impairing well-being, increasing suicidality risk, and imposing a substantial economic burden, estimated at up to $6.5 trillion annually, due largely to productivity losses [9, 12, 13]. Despite this prevalence, most software systems inadequately support anxiety-prone users, even though digital interactions can influence mental health [3]. Interventions, such as TikTok restricted beauty filters for users under 18 due to concerns about psychological harm and unrealistic beauty standards [15], remain limited.

With the growing immersiveness and pervasiveness of information and communication technologies (ICT), the absence of anxiety-aware design principles risks exacerbating or inducing anxiety, while current software engineering practices typically address such concerns only when anxiety-prone users are explicitly defined as a target group in digital health applications.

Requirements engineering (RE), a critical determinant of software project success, has increasingly incorporated emotional considerations [7]. However, these efforts do not adequately address the specific needs of anxiety-prone end-users. Existing research on emotional requirements often emphasizes eliciting or enhancing positive emotions (e.g., joy or excitement) while overlooking the prevention or mitigation of negative emotions such as anxiety [1, 8]. For example, emotional intensity maps are used to shape player experiences without targeting anxiety reduction [4], and platforms like MemoRE focus on recognizing emotions such as joy, sadness, or fear but do not explicitly address anxiety or stress [5].

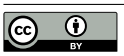
Research on computer anxiety (computerphobia or cyberphobia) primarily examines fear or discomfort in specific groups, such as older adults or technology novices [10]. In contrast, this vision paper targets individuals with general anxiety disorders [11] who use everyday digital systems and proposes an anxiety-aware requirements engineering (AWARE) framework that integrates anxiety-related principles into RE through a quality-in-use perspective.

The ISO/IEC 25019:2023 standard defines quality in use through three characteristics: (1) beneficialness, encompassing usability, accessibility, and suitability; (2) freedom from risk, including economic, environmental and societal, health, and human life risks; and (3) acceptability, comprising experience, trustworthiness, and compliance. While accessibility for physical impairments is well established, mental health remains largely overlooked [3]. In particular, freedom from risk lacks attributes or metrics addressing anxiety, as health risk is defined without explicit consideration of mental health, leading many software engineers to overlook its relevance when defining requirements, designing systems, and assessing the impact on anxiety-prone users.

2 The approach: AWARE theoretical framework

Fig. 1 presents an overview of the AWARE theoretical framework, which comprises three key RE phases, elicitation, analysis, and specification, aimed at ensuring that software requirements address the needs of anxiety-prone users.

Relevant stakeholders. According to the ISO/IEC 25030:2019 standard, software stakeholders include primary, secondary, and indirect users, as well as developers, evaluators, and society, with users serving as key sources of quality-in-use requirements. We argue that anxiety-prone users, particularly those with mild to moderate anxiety, should be recognized as primary stakeholders, even when software is not explicitly designed for anxiety-related purposes. Although they may not directly elicit requirements, their needs and experiences must be considered due to the potential for emotional harm, underscoring the importance of integrating anxiety-prone users into stakeholder analysis.



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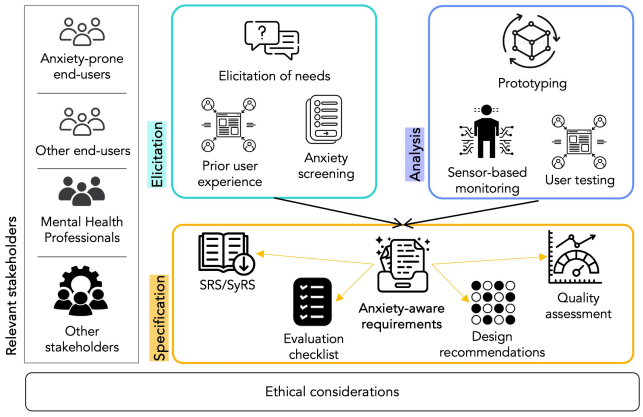


Figure 1: Anxiety-aware RE framework

Elicitation. The elicitation phase aims to capture stakeholder needs, including those of anxiety-prone users, which traditional methods (e.g., interviews, surveys) may fail to address adequately. We therefore propose a two-part questionnaire: (1) anxiety screening using validated instruments such as GAD-7 or PHQ-9 to identify users most affected by anxiety in digital interactions [16, 17]; and (2) prior user experience based on mental health oriented accessibility guidelines, focusing on cognitive load, user control, and avoidance of visually overwhelming stimuli [3].

Analysis. The analysis phase assesses the impact of prototype design on user anxiety using sensor-based testing, with iterative prototyping enabling the mitigation of anxiety-triggering elements. Because conventional usability evaluations often miss emotional distress, the framework integrates biometric and behavioral sensing to detect physiological indicators of anxiety, such as irregular heartbeat, muscle tension, and breathing difficulties [6, 11]. These technologies, widely used in biofeedback interventions, provide real-time insights for anxiety management and are increasingly accessible and cost-effective [2, 14]. It should be noted that sensors involving extensive wiring or intrusive equipment may increase anxiety during the analysis phase. Therefore, we recommend using devices that anxiety-prone individuals can wear comfortably, such as heart rate monitors via smartwatches and eye-tracking glasses.

Specification. The specification phase integrates insights from the elicitation and analysis phases into defining anxiety-aware requirements. These should be documented and should inform evaluation checklists, design recommendations, and quality assessments to ensure user control over stress-inducing interactions. To ensure rigor and consistency, the specified requirements should conform to the quality criteria defined in ISO/IEC/IEEE 29148:2018, which sets out quality criteria such as clarity, consistency, feasibility, verifiability, and traceability.

Example. As an illustrative example, the proposed AWARE framework can be applied to the design of an e-commerce platform, where features such as time-limited deals, overwhelming choice, excessive notifications, and complex checkout processes often contribute to stress and decision fatigue. Through the framework's phases, the platform can be refined to reduce anxiety triggers and translate insights into concrete requirements, such as a calming interface mode or transparent delivery and return information.

3 Current work and future plans

We are currently conducting a literature review to investigate existing approaches for reducing anxiety and stress in software systems. Following ethical approval, we will collaborate with mental health professionals and individuals with mild anxiety to elicit context-specific requirements for web-based applications. These insights will inform a reusable catalogue of anxiety-aware requirements to support practitioners, researchers, and artificial intelligence (AI)-assisted RE. Finally, we will empirically evaluate the approach using the proposed analysis methods, with the goal of making anxiety awareness a practical part of RE and software engineering practice.

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